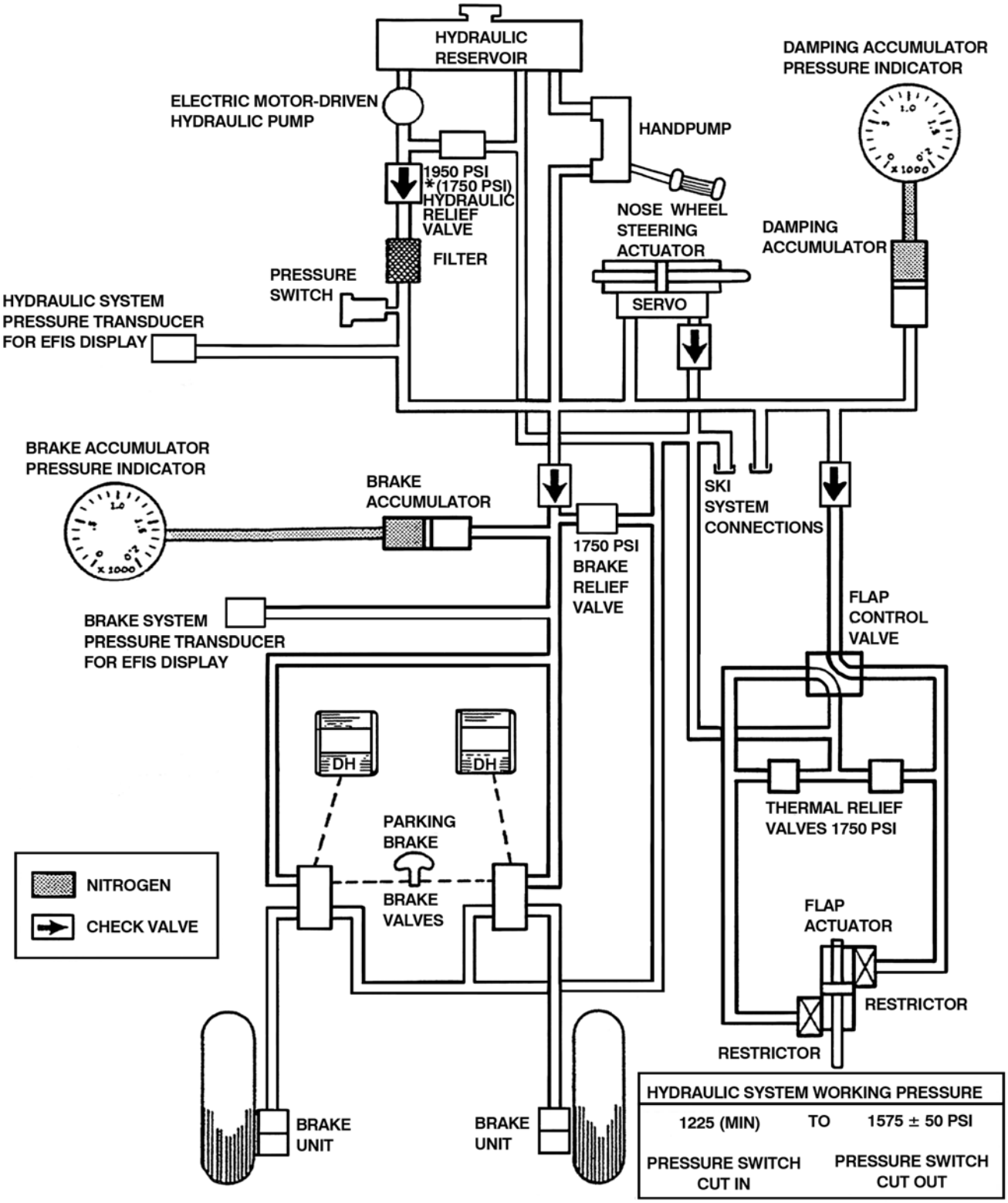
Hydraulic Power – General Data

1. General

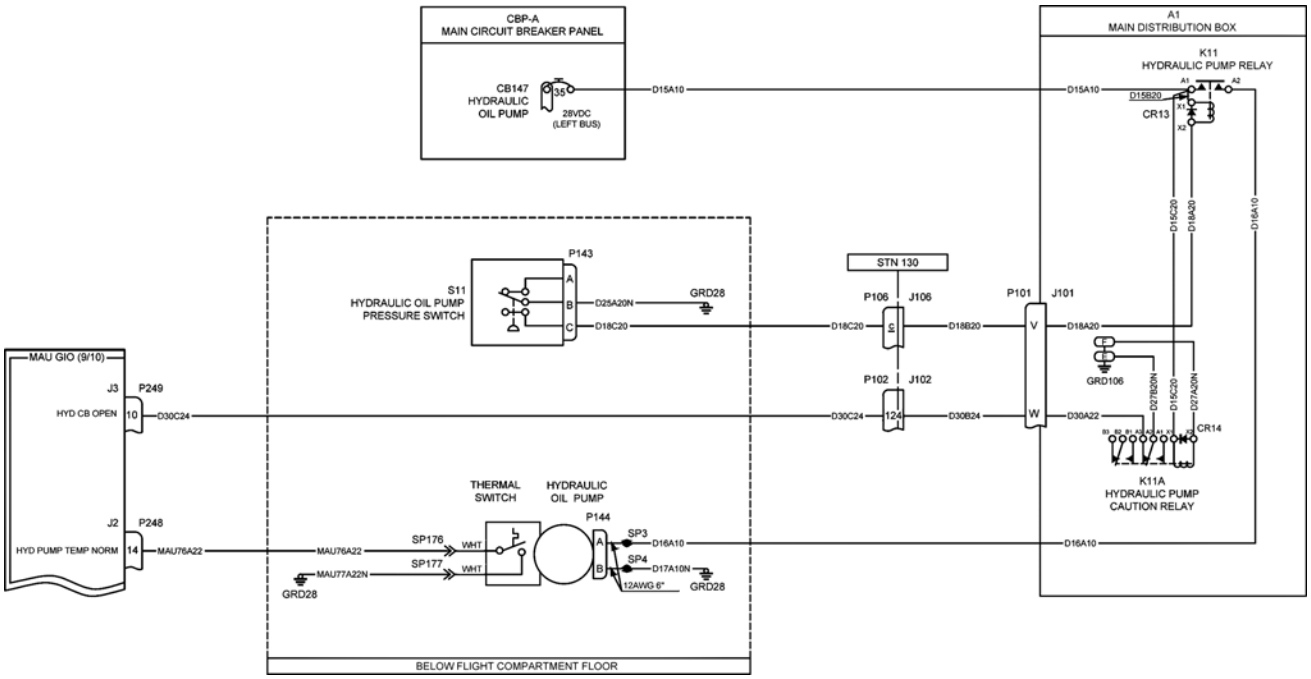
The hydraulic system supplies power to operate the wing flaps, wheel brakes and nose wheel steering. System pressure is supplied by an electric motor-driven hydraulic pump. Secondary sources of hydraulic power are: two hydraulic accumulators, one for airframe system pressure, the other for brakes system, and a hydraulic handpump. Flared pipeline couplings are used throughout the system. Refer to Figure 1 and Figure 2 for details.

Hydraulic system power is supplied by the electric motor-driven hydraulic pump which is supplied with fluid from a reservoir through a suction line, and delivers fluid through a system filter to the various hydraulic circuits. A working pressure of approximately 1575 psi is regulated by a pressure switch, which controls switching the pump motor on and off. An airframe system relief valve relieves any pressure buildup in excess of 1950 psi at full flow, and two thermal relief valves set at 1750 psi cracking pressure protect the flap circuit. A pressure relief valve, also of 1750 psi cracking pressure, is installed in the wheel brakes circuit. Fluid from all circuits is returned to the reservoir through a common return line. The airframe hydraulic system pressure and brake system pressure are indicated on the Engine Indication and Crew Alerting System (CAS) display. The main components of the hydraulic system are mounted together on a hydraulic power package below the flight compartment floor.

Electrical power to operate the electric motor-driven hydraulic pump, is derived from a 35-ampere circuit breaker and the contacts of a hydraulic pump relay, when the relay is energized. Because the relay control coil obtains a ground through the pressure switch contacts, the relay is energized only when the system hydraulic pressure falls below the working pressure value. The circuit breaker, labeled HYDRAULIC OIL PUMP, is located on the main circuit breaker panel, and the relay is mounted in the main distribution box. A hydraulic pump circuit breaker open warning is provided. A relay, mounted in the main electrical distribution box is normally energized when the HYDRAULIC OIL PUMP circuit breaker is engaged. A temperature switch on the pump housing triggers a CAS caution “Hyd Pump Over Temp”.



Hydraulic System Schematic
Figure 1



Hydraulic System Electrical – Schematic
Figure 2

2. Servicing

A. Hydraulic Lines

Removal and installation of hydraulic lines is self-evident, but the following points must be noted:

1) Before removing a pressure line, or a component in the pressure lines, discharge hydraulic pressure by operating wing flaps (for damping accumulator), or pump brake pedals (for brake accumulator).

2) Before removing an accumulator, air pressure lines, indicator or charging valve, discharge all air pressure by loosening air valve approximately one-quarter turn and waiting until all air has escaped.

WARNING **FAILURE TO PROPERLY DISCHARGE ALL AIR PRESSURE FROM AN ACCUMULATOR. MAY RESULT IN SERIOUS INJURY TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.**

3) Cap all openings on disconnection of lines. Openings must remain blanked until immediately before reconnection.

4) For torque loading procedures of fluid lines and fittings, refer to 20-20-01, Fluid Lines and Fittings – Torquing.

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